

Brenton Law

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EDUCATION

University of California, Los Angeles

Expected Aug 2026

M.Eng. Artificial Intelligence

University of California, Berkeley

May 2022

B.A. Applied Mathematics, B.A. Statistics

GPA: 3.92

EXPERIENCE

Optiver

Chicago, IL

Trading Operations Analyst

Jul 2023 – Aug 2024

- Spearheaded automation initiatives by developing a Python-based pricing pipeline, streamlining market entry
- Engineered tooling leveraging PostgreSQL to optimize quoting for new expiries, ensuring market edge capture
- Designed and implemented a robust data pipeline for parsing execution logs into S3, enabling proactive identification of application performance issues and configuration anomalies
- Assessed third-party data efficacy through statistical analysis, enhancing accuracy in predicting true company event timings, forecasting and generating summary statistics for over 70 symbols
- Conducted research in C on production simulations of systematic trading applications, backtesting to improve order success rates and maximize profits in real world environments

Quantitative Trader

Aug 2022 – Jul 2023

- Traded on the Single Stock Options financials desk and Asian Trading Hours desk both on-screen and with brokers
- Worked on projects investigating efficacy of Order Insert Retreats in illiquid markets, built revamped KPI framework for Global Trading Hours, and writing a script to properly align settlement calendars

Belvedere Trading

Chicago, IL

Junior Trading Intern

Jul 2021 – Aug 2021

- Studied options theory coursework with on-desk assignments and exams, scoring in top three of intern class
- Identified relationships between the firm's edge retention and market volatility via statistical models and visualizations
- Worked in JupyterHub on projects including pulling edge retention statistics with greeks to inform real-time trading decisions and algorithmically identifying repeat order flow

RESEARCH

UC Berkeley Statistics Department

Berkeley, CA

Undergraduate Research Assistant

May 2020 – Feb 2021

- Used Matlab to implement novel RODE and SDE methods for one- and two-dimensional applied neuron problems with colored noise by using lasso regression to estimate density functions
- Wrote convergence study scripts to evaluate speeds and accuracies for solutions of various problem formulations
- Optimized application of algorithms by testing various approaches to speeding up performance of computation involving fine discretization meshes and large spatial-temporal matrices

PROJECTS

Sports Betting

- Utilizes LightGBM regressor in Python to predict spreads of college football games from historical data
- Streamlined feature development using scikit-learn pipelines and transformers, improved through backtesting predictive and PnL metrics

LoL Main Recommender

- Constructed recommendation system in Python for champions in the video game League of Legends
- Implemented various filters in PyTorch, including collaborative, content-based, and two-tower

SKILLS

Languages & Tools: Python (pandas, NumPy, scikit-learn, PyTorch), Git, SQL (Postgres), Matlab, R